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# Effect of Sampling and Image Formation on Object Visibility
# Google Colab
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

image_name = list(uploaded.keys())[0]

# Read image
img = cv2.imread(image_name)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# 1. Low Sampling (Low Resolution)
# -----

small = cv2.resize(img,
                    (img.shape[1]//6, img.shape[0]//6),
                    interpolation=cv2.INTER_AREA)

low_resolution = cv2.resize(small,
                             (img.shape[1], img.shape[0]),
                             interpolation=cv2.INTER_NEAREST)

# -----
# 2. Poor Image Formation (Blur)
# -----

blurred = cv2.GaussianBlur(low_resolution, (11,11), 5)

# -----
# 3. Image Enhancement
# -----

enhanced = cv2.detailEnhance(blurred,
                             sigma_s=10,
                             sigma_r=0.15)

# -----
# Display Results
# -----
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plt.figure(figsize=(15,8))

plt.subplot(2,2,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(2,2,2)
plt.imshow(low_resolution)
plt.title("Low Resolution (Low Sampling)")
plt.axis("off")

plt.subplot(2,2,3)
plt.imshow(blurred)
plt.title("Blurred Image (Poor Image Formation)")
plt.axis("off")

plt.subplot(2,2,4)
plt.imshow(enhanced)
plt.title("Enhanced Image")
plt.axis("off")

plt.tight_layout()
plt.show()

# -----
# Explanation
# -----

print("\nObject Visibility Analysis")
print("-----")
print("Low Sampling:")
print("- Fewer pixels capture less detail.")
print("- Small objects become pixelated or disappear.")

print("\nPoor Image Formation:")
print("- Blur reduces edge sharpness.")
print("- Fine features become difficult to distinguish.")

print("\nImprovement:")
print("- Capture images at higher resolution.")
print("- Use a good camera and proper focus.")
print("- Apply image enhancement techniques to improve clarity.")
```



Choose Files number_plate.jpg

number_plate.jpg(image/jpeg) - 14891 bytes, last modified: 7/13/2026 - 100% done

Saving number_plate.jpg to number_plate.jpg

Original Image



Low Resolution (Low Sampling)

